FACIAL OCCLUSION DETECTION AND CLASSIFICATION IN CONTROLLED LABORATORY ENVIRONMENTS

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Abstract

Laboratory environments require appropriate personal protective equipment (PPE) and reliable monitoring of safety practices. Masks, goggles, and other facial coverings can complicate automated visual assessment. This paper proposes a computer-vision framework for facial occlusion detection and classification in controlled laboratory environments. OpenCV is used to acquire video frames, and YuNet is used to detect faces. The detected facial regions are then processed by a convolutional neural network (CNN) to classify the observed condition into three categories: Fully Visible, Safety PPE Compliant, and Unauthorized Occlusion. Safety PPE Compliant refers to a visible facial condition in which a face mask and protective eyewear are present, while Unauthorized Occlusion refers to facial obstruction that does not meet the project's predefined PPE criteria. Facial key-point and covered-region analysis are intended to support the identification of occlusion around the eyes, nose, and mouth. The system is designed to compare predictions with predefined laboratory safety rules and support alerts and event logging, including timestamps, predicted classes, and snapshots, for review. The planned dataset is intended to include variation in lighting, face angle, distance, skin tone, and degree of occlusion. The framework aims to support more consistent visual monitoring; however, classification performance and processing latency must be measured on a suitable held-out test set before claims about accuracy or real-time performance can be established.

Keywords— Facial Occlusion; PPE Detection; Deep Learning; Convolutional Neural Network; Computer Vision; Laboratory Safety; OpenCV; YuNet

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